Offline preparation (fixed before testing)

Chronological training

Calibration and cutoff development

Fixed electrical model, sensitivities, tariffs and gate settings

Seasonal chronological evaluation (context only)

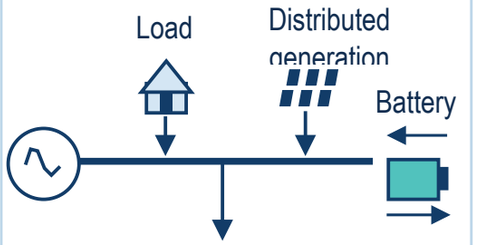
Frozen predictor

Joint load-generation residual half-widths q

Fixed electrical settings

Panel A: Measurements and state

Distribution network and battery



Primary observations $z(t)$ — Timestamp age $a(t)$ availability $m(t)$

Forward carry for missing readings

Current, lag-1 and lag-24 features + calendar

Reference r(t) with availability flag

Battery state of charge $x(t)$

Panel B: Forecast and checked proposal

Frozen random forest

Forecast $\hat{y}(t+1|t)$
Operating / profile factors

Past-prediction buffer

Nine battery candidate levels

-1, -0.75, -0.50, -0.25, 0, 0.25, 0.50, 0.75, 1 × rated power

Screen battery bounds → **Rank by local AC approximation + operating cost** → **Exact AC check of candidates**

Checked proposal $u^*(t)$

No feasible candidate: checked zero; feasibility flag retained

Panel C: Causal evidence and dispatch regime

Completed-horizon buffer
Completed performance only

$\hat{y}(t|t-1)$

r(t)

A prediction made at t is compared with $r(t+1)$ only at the next hour.

t+1

F: freshness	C: completeness	M: model performance	A: meter agreement
exp[-a/2]	Latest 6 observations	Latest 24 completed pairs	Primary vs reference

Evidence score
$T = 0.30F + 0.25C + 0.25M + 0.20A$

Full: $T \geq \tau H$	Cautious: $\tau L \leq T < \tau H$	Hold: $T < \tau L$
Command: u^*	Command: $0.5u^*$	Command: 0

Reference outage: hold
Score is operating evidence

Panel D: Command validation and execution

Checked proposal $u^*(t)$ (from Panel B)

Dispatch regime (from Panel C)

Apply dispatch regime
Select full, half-power or zero command

Exact AC check of actual command at forecast state

Non-zero command passes?

No → **Replace by zero; check and log hold** (feasibility flag retained)

Yes → **Issue battery command** $u(t)$ (with feasibility flag)

Zero command

Update battery state from issued power

Next decision: battery state x(t+1)

Next-hour plant state available after command issue

Independent realised-outcome evaluation
(no feedback to decision)

Issued command $u(t)$

Realised load-generation state $y(t+1)$

Realised AC power flow

Voltage and thermal feasibility

Energy-adjusted cost and network loss

Violation severity, duration and retained control activity

Current-decision information
Past / next-decision feedback
Frozen offline parameters
A B C D Panels

Next decision: add completed pair $(\hat{y}(t+1|t), r(t+1))$ to buffer